

## Education

### University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

**Relevant Coursework:** Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

## Skills

**Languages:** Python, TypeScript, C++, SQL, Java

**Software:** Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

## Experience

### L3Harris Technologies

Associate Software Engineer

Dallas, TX

June 2025 – Present

- Developed signal processing modules in Python and C++ on embedded systems, collaborating with hardware teams to optimize performance and resource utilization.
- Built microservices for high-frequency data ingestion, processing, and storage, leveraging Podman and serverless functions.
- Automated CI/CD testing with pipelines, integrating static code analysis and integration testing.

### Doxy.me

Software Engineer

Charleston, SC

August 2024 – May 2025

- Built data-intensive internal applications using Python backend APIs and TypeScript frontend components, integrating real-time clinical data ingested from WebRTC video sessions with LLM-powered extraction pipelines (PyTorch, Llama) deployed on AWS SageMaker and Fargate.
- Improved data reliability by designing REST API services backed by relational data models, normalizing heterogeneous inputs from third-party clinical systems into consistent schemas and persisting structured outputs for downstream analytics and reporting.
- Maintained high test coverage by authoring unit and integration tests across backend services and frontend workflows, participating in code reviews, and owning CI/CD configuration on AWS to ensure stable, reproducible deployments.

### HEXstream

Software Engineering Intern

Chicago, IL

May 2022 – Aug 2022

- Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.
- Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.

## Projects

**University Course Planning/Review Platform (Class Best):** Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

**Telehealth Mental Health Diagnosis Tool (Doxy.me):** Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

**Configuration Fuzzer:** Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.